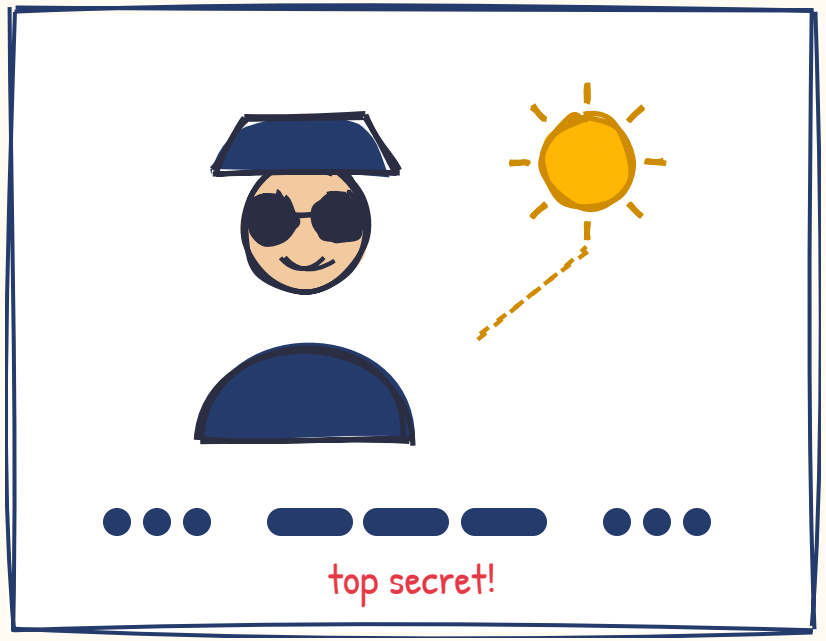
Little Explorers · Class 12

Morse Code Light Messenger

Secret Agent training starts today!

Dots and dashes **Button + light + beep** **Sending messages**



Welcome, Agents!
Today you learn a secret code that ships, pilots and spies have used for over 180 years.

Agent challenge

Can you talk without words?

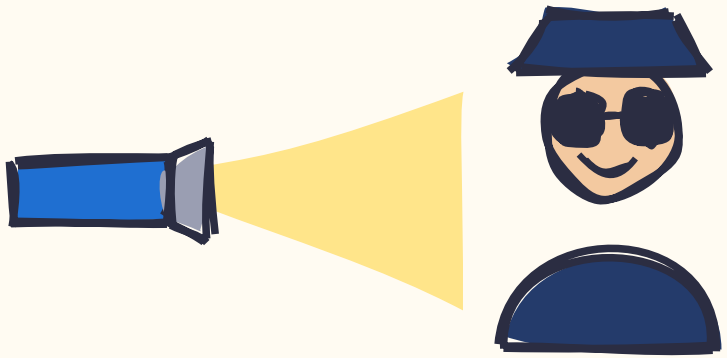
Your friend is far across the playground. It is too noisy to shout.

You only have a **torch**. How can you send a message?

Wave it around?

Draw in the air?

Short and long flashes!



flash... flash... flash

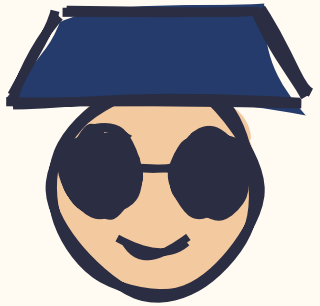
Watch the teacher

The teacher flashes a torch: short, short, short. What do you think it means?

LEARN

Agent training goals

Today, I will learn to...



1 I can tell a **dot** (short) from a **dash** (long).

2 I can send **SOS** and my **initials** with a button, a light and a beep.

3 I can **decode** a message from my partner using a chart.

Keywords

code · dot · dash · message · send · decode · SOS

Watch and think

Video time!



Make Your Own Morse Code (for kids)

ZeeKay Education · about 5 minutes · youtu.be/KXt0LK3MMm8

Watch for...

1. What are the two signals in Morse code?
2. How can you send Morse code with a **light**?
3. Why did people use Morse code long ago?

Scan to watch

After the video: Flash your partner's first letter with your hands: open = long, quick blink = short.

The secret

Just two signals: short and long

DOT



a **short** flash or beep

"dit!"

DASH



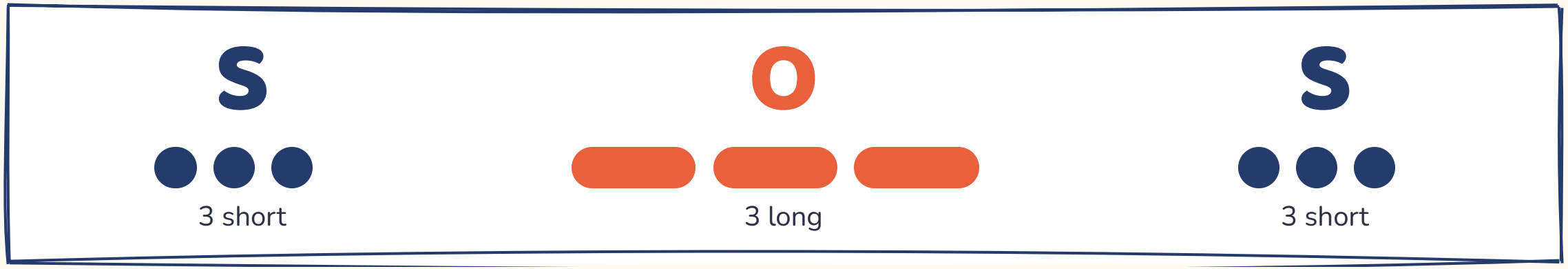
a **long** flash or beep

"daaah!"

A dash is as long as **3 dots**. Count it: dit = "1", daaah = "1, 2, 3".

The most famous message

S O S: please help!



Ships in trouble sent **SOS**. It was chosen because it is **easy to send** and **hard to mistake**.

Let's try!
Clap SOS together: clap clap clap, rub rub rub, clap clap clap!

Agent training

Practise sending

Mission 1: send SOS



Tap tap tap, hold hold hold, tap tap tap.

Agent tips

Count in your head: dit = 1, daaah = 1, 2, 3.

Wait a little between letters.

Go **slowly**. Fast is not better!

Mission 2: send your initials

Find your first letter on the codebook. Practise it 3 times.

Example: **A** for Ahmed



Partner check

Your partner watches the light. Can they tell which letter you sent?

Look around you

Secret patterns are everywhere!



Wi-Fi
sends patterns through the air



TV remote
flashes invisible light patterns



Phones
turn your voice into patterns



Computers
think in just two signals: 0 and 1



Morse uses **2 signals** (dot and dash). Computers also use **2 signals** (0 and 1). You just learned how computers talk!

Wow!
Your phone's camera can see a TV remote's secret flashes. Try it at home!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

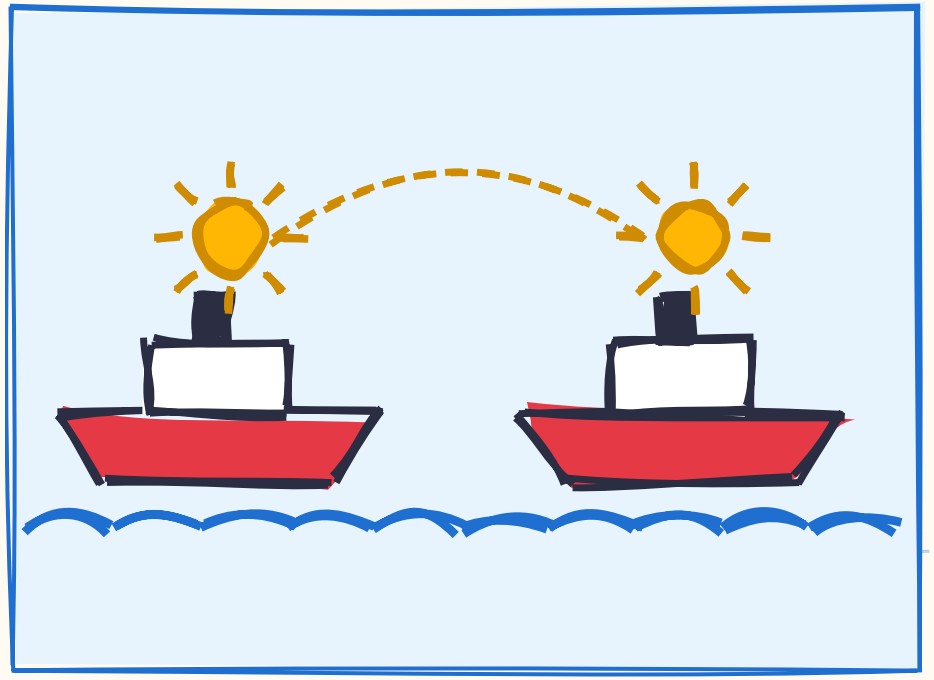
1. Before phones, there was Morse code
2. Be the Morse code!
3. Our first secret letters
4. The full alphabet (find your initials!)
5. Meet the Light Messenger
6. Let's code our messenger
7. Send and decode a message

A long time ago...

Before phones, there was Morse code

No phones, no internet!

About 180 years ago, **Samuel Morse** and **Alfred Vail** made a code of **beeps**. Messages zoomed along wires from city to city. It was called the **telegraph**.



Ships at sea used **flashing lamps** to talk to each other, even in the dark!

Still used today!
Pilots and radio fans still learn Morse code.

Stand up and play!

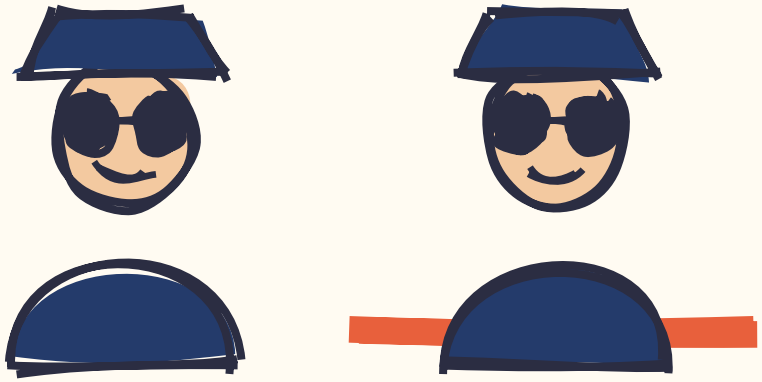
Be the Morse code!

How to play

Teacher says **"dit"**: jump once on the spot!

Teacher says **"daaah"**: stretch your arms out wide, slowly!

Now the teacher **claps** instead. Short clap or long rub?



dit = jump **daaah = stretch**

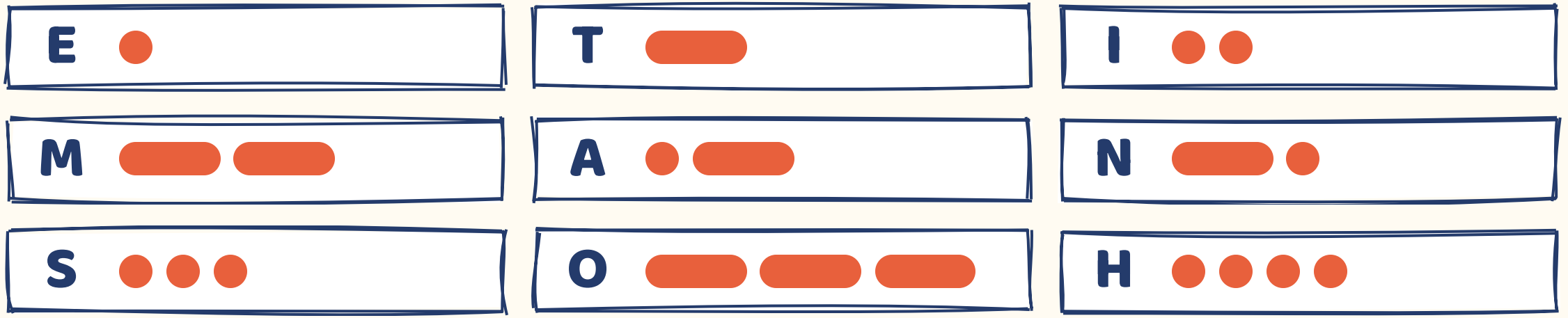
Round 2: the teacher plays a pattern. Copy it back as a class! dit dit daaah...

Quiet please

Agents listen carefully. Silence between signals matters too!

Agent codebook

Our first secret letters



Look for patterns: **E** is 1 dot, **I** is 2 dots, **S** is 3 dots, **H** is 4 dots!

Agent codebook

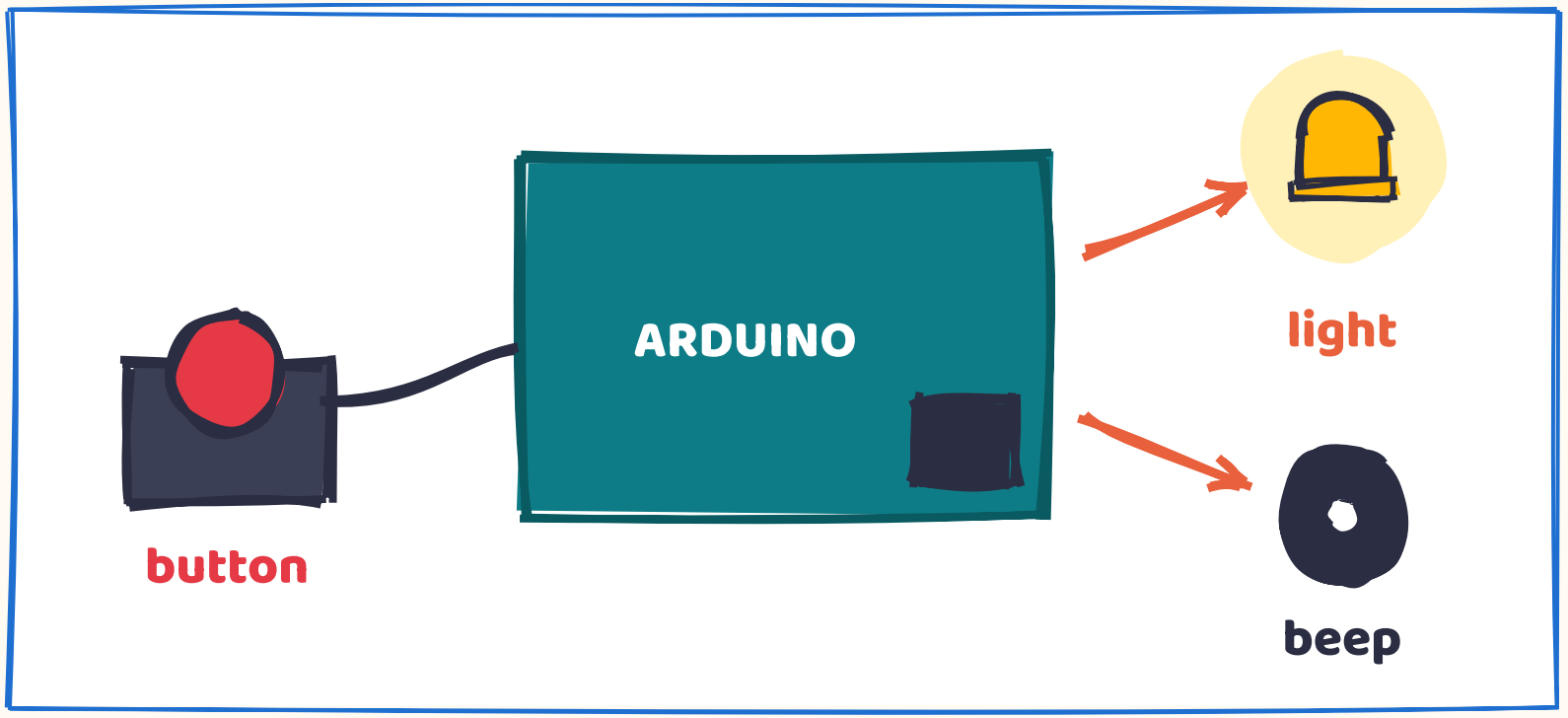
The full alphabet (find your initials!)

A 	B 	C 	D 
E 	F 	G 	H 
I 	J 	K 	L 
M 	N 	O 	P 
Q 	R 	S 	T 
U 	V 	W 	X 
Y 	Z 		

Teacher: print this page as the decoding chart for each pair.

Our builder's kit

Meet the Light Messenger



How it works

Press the button: light **ON** and beep!

Let go: light **OFF**, quiet.

Short press = **dot**.

Long press = **dash**.

Teacher: the button, LED and buzzer are pre-wired. Children plug in, code and test.

Input (button) → brain (Arduino) → outputs (light and beep). Same pattern as every machine we have built!

MAKE

Time to build!

Let's code our messenger

```
when Arduino starts
forever
    if button is pressed
        turn ON light
        turn ON beep
    else
        turn OFF light
        turn OFF beep
```

3 easy steps

1. Snap the blocks
2. Upload
3. Press the button: short, then long!



Remember IF and ELSE?
IF the button is pressed, light ON. ELSE light OFF. Just like our night lamp!

Secret mission!

Send and decode a message

How it works (pairs)

Sender: pick a secret mission card. Do not show it! Send it with the button.

Decoder: draw a dot or a dash for each signal. Use the codebook to find the word.

Then **swap jobs**!

Mission card 1

HI

•••• ••

Mission card 2

SOS

••• — — — •••

Mission card 3

MOM

— — — — — —

Mission card 4

SUN

••• •• — — •

Decoder sheet: _ _ _ _ / _ _ _ _ / _ _ _ _

Top secret

Teacher: print and cut out the mission cards.

Finished early?

Super agent challenges

Challenge time!



Across the room: send a secret word to another group far away. Can they read your light?



Light only: switch off the beep. Can your partner decode with just the light?



Robot sender: use blocks to make the Arduino flash SOS all by itself!

Robot sender blocks

repeat 3: light ON 0.3 s, OFF 0.3 s

repeat 3: light ON 1 s, OFF 0.3 s

repeat 3: light ON 0.3 s, OFF 0.3 s

Think!
Why do we need a little gap between each flash?

Show what you know

What did we learn?

- [x] Morse code uses dots (short) and dashes (long)
- [x] SOS = 3 short, 3 long, 3 short
- [x] Light and sound can carry messages
- [x] A code is a pattern that both people understand



Secret Agent
badge unlocked!

At home
Teach your family SOS. Tap it on the table at dinner!

Exit ticket: flash or clap SOS for the teacher on your way out!

Next class: a Makey Makey Fruit Piano!